

Approved Sample Lesson Collection

Eight sample lessons across computing, mathematics, science, and communication.

Reference material for lesson planning and course-content development.

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1. Approved sample: Python Functions

Subject: Computing. Level: Beginner. Suggested duration: 60 minutes.

Lesson goal: Learners define and call a function, pass one argument, return a value, and explain the difference between print and return.

Learning objectives

- Explain the main idea in Python Functions using the terms introduced in the lesson.
- Complete a short worked or guided task.
- Complete one independent task and explain one decision.

Opening

Begin with a short example that makes the need for Python Functions visible. Ask one diagnostic question before giving the formal rule. Keep the opening to roughly five minutes so the main lesson has enough time.

Teaching sequence

- Why functions are useful
- Defining and calling
- Parameters and arguments
- Return values
- Short debugging example

The first worked example should be fully explained. The second should remove one layer of support and ask the learner to predict the next step or result.

Practice

- Predict the return value
- Fix a function that prints instead of returning
- Write a temperature-conversion function

Quick check

Use four to six short questions. At least one should require reasoning rather than a definition. Include a short answer key with explanations for likely misconceptions.

Teacher notes

If learners struggle, return to the smallest example and make the intermediate state visible. If learners finish quickly, use a new context that requires the same core idea rather than introducing an unrelated topic.

Quality notes

The lesson is considered complete when the objective, examples, practice, and assessment all point to the same skill. Avoid adding extra content just because related material is available in the reference library.

2. Approved sample: SQL JOIN and GROUP BY

Subject: Computing. Level: Intermediate. Suggested duration: 75 minutes.

Lesson goal: Learners connect users and orders, write a grouped count, and explain why a wrong JOIN condition can produce believable data.

Learning objectives

- Explain the main idea in SQL JOIN and GROUP BY using the terms introduced in the lesson.
- Complete a short worked or guided task.
- Complete one independent task and explain one decision.

Opening

Begin with a short example that makes the need for SQL JOIN and GROUP BY visible. Ask one diagnostic question before giving the formal rule. Keep the opening to roughly five minutes so the main lesson has enough time.

Teaching sequence

- Review SELECT and WHERE
- Relationship between tables
- INNER JOIN
- GROUP BY with COUNT/SUM
- Wrong JOIN diagnosis

The first worked example should be fully explained. The second should remove one layer of support and ask the learner to predict the next step or result.

Practice

- Predict joined rows
- Fix a JOIN condition
- Write total order value per user

Quick check

Use four to six short questions. At least one should require reasoning rather than a definition. Include a short answer key with explanations for likely misconceptions.

Teacher notes

If learners struggle, return to the smallest example and make the intermediate state visible. If learners finish quickly, use a new context that requires the same core idea rather than introducing an unrelated topic.

Quality notes

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3. Approved sample: Linear Equations

Subject: Mathematics. Level: Beginner. Suggested duration: 60 minutes.

Lesson goal: Learners solve one-step and two-step equations and check a solution by substitution.

Learning objectives

- Explain the main idea in Linear Equations using the terms introduced in the lesson.
- Complete a short worked or guided task.
- Complete one independent task and explain one decision.

Opening

Begin with a short example that makes the need for Linear Equations visible. Ask one diagnostic question before giving the formal rule. Keep the opening to roughly five minutes so the main lesson has enough time.

Teaching sequence

- Equality idea
- Inverse operations
- One-step equations
- Two-step equations
- Checking by substitution

The first worked example should be fully explained. The second should remove one layer of support and ask the learner to predict the next step or result.

Practice

- Choose next valid step
- Find and fix a sign error
- Solve and check one equation

Quick check

Use four to six short questions. At least one should require reasoning rather than a definition. Include a short answer key with explanations for likely misconceptions.

Teacher notes

If learners struggle, return to the smallest example and make the intermediate state visible. If learners finish quickly, use a new context that requires the same core idea rather than introducing an unrelated topic.

Quality notes

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4. Approved sample: Motion and Speed

Subject: Physics. Level: Beginner. Suggested duration: 75 minutes.

Lesson goal: Learners distinguish distance and displacement, calculate average speed, and interpret a simple motion table.

Learning objectives

- Explain the main idea in Motion and Speed using the terms introduced in the lesson.
- Complete a short worked or guided task.
- Complete one independent task and explain one decision.

Opening

Begin with a short example that makes the need for Motion and Speed visible. Ask one diagnostic question before giving the formal rule. Keep the opening to roughly five minutes so the main lesson has enough time.

Teaching sequence

- Distance vs displacement
- Average speed
- Units
- Round-trip example
- Simple graph interpretation

The first worked example should be fully explained. The second should remove one layer of support and ask the learner to predict the next step or result.

Practice

- Calculate average speed
- Explain zero displacement on a round trip
- Identify a units error

Quick check

Use four to six short questions. At least one should require reasoning rather than a definition. Include a short answer key with explanations for likely misconceptions.

Teacher notes

If learners struggle, return to the smallest example and make the intermediate state visible. If learners finish quickly, use a new context that requires the same core idea rather than introducing an unrelated topic.

Quality notes

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5. Approved sample: Cell Structure

Subject: Biology. Level: Beginner. Suggested duration: 60 minutes.

Lesson goal: Learners identify major cell structures and connect each structure to its function.

Learning objectives

- Explain the main idea in Cell Structure using the terms introduced in the lesson.
- Complete a short worked or guided task.
- Complete one independent task and explain one decision.

Opening

Begin with a short example that makes the need for Cell Structure visible. Ask one diagnostic question before giving the formal rule. Keep the opening to roughly five minutes so the main lesson has enough time.

Teaching sequence

- Cell membrane
- Cytoplasm
- Nucleus
- Mitochondria
- Plant and animal cell differences

The first worked example should be fully explained. The second should remove one layer of support and ask the learner to predict the next step or result.

Practice

- Match structure to function
- Correct a misconception about mitochondria
- Compare plant and animal cells

Quick check

Use four to six short questions. At least one should require reasoning rather than a definition. Include a short answer key with explanations for likely misconceptions.

Teacher notes

If learners struggle, return to the smallest example and make the intermediate state visible. If learners finish quickly, use a new context that requires the same core idea rather than introducing an unrelated topic.

Quality notes

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6. Approved sample: Acids, Bases, and pH

Subject: Chemistry. Level: Intermediate. Suggested duration: 60 minutes.

Lesson goal: Learners interpret pH values, describe indicators, and explain a simple neutralization example.

Learning objectives

- Explain the main idea in Acids, Bases, and pH using the terms introduced in the lesson.
- Complete a short worked or guided task.
- Complete one independent task and explain one decision.

Opening

Begin with a short example that makes the need for Acids, Bases, and pH visible. Ask one diagnostic question before giving the formal rule. Keep the opening to roughly five minutes so the main lesson has enough time.

Teaching sequence

- pH scale
- Logarithmic idea at a basic level
- Indicators
- Neutralization
- Safety

The first worked example should be fully explained. The second should remove one layer of support and ask the learner to predict the next step or result.

Practice

- Classify pH values
- Interpret indicator results
- Explain why changing a coefficient is different from changing a formula

Quick check

Use four to six short questions. At least one should require reasoning rather than a definition. Include a short answer key with explanations for likely misconceptions.

Teacher notes

If learners struggle, return to the smallest example and make the intermediate state visible. If learners finish quickly, use a new context that requires the same core idea rather than introducing an unrelated topic.

Quality notes

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7. Approved sample: Professional Email Writing

Subject: Business Communication. Level: Beginner. Suggested duration: 45 minutes.

Lesson goal: Learners write a short email with a clear purpose, useful context, and a specific next step.

Learning objectives

- Explain the main idea in Professional Email Writing using the terms introduced in the lesson.
- Complete a short worked or guided task.
- Complete one independent task and explain one decision.

Opening

Begin with a short example that makes the need for Professional Email Writing visible. Ask one diagnostic question before giving the formal rule. Keep the opening to roughly five minutes so the main lesson has enough time.

Teaching sequence

- Subject line
- Opening purpose
- Necessary context
- Specific request
- Professional tone

The first worked example should be fully explained. The second should remove one layer of support and ask the learner to predict the next step or result.

Practice

- Improve an unclear subject line
- Rewrite a vague request
- Draft a short follow-up email

Quick check

Use four to six short questions. At least one should require reasoning rather than a definition. Include a short answer key with explanations for likely misconceptions.

Teacher notes

If learners struggle, return to the smallest example and make the intermediate state visible. If learners finish quickly, use a new context that requires the same core idea rather than introducing an unrelated topic.

Quality notes

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8. Approved sample: Reading Charts and Tables

Subject: Statistics. Level: Beginner. Suggested duration: 60 minutes.

Lesson goal: Learners read exact values, compare categories, identify a trend, and notice a misleading axis.

Learning objectives

- Explain the main idea in Reading Charts and Tables using the terms introduced in the lesson.
- Complete a short worked or guided task.
- Complete one independent task and explain one decision.

Opening

Begin with a short example that makes the need for Reading Charts and Tables visible. Ask one diagnostic question before giving the formal rule. Keep the opening to roughly five minutes so the main lesson has enough time.

Teaching sequence

- Bar charts
- Line charts
- Table lookup
- Trend statements
- Misleading scales

The first worked example should be fully explained. The second should remove one layer of support and ask the learner to predict the next step or result.

Practice

- Read a value
- Write one supported conclusion
- Identify why a truncated axis can mislead

Quick check

Use four to six short questions. At least one should require reasoning rather than a definition. Include a short answer key with explanations for likely misconceptions.

Teacher notes

If learners struggle, return to the smallest example and make the intermediate state visible. If learners finish quickly, use a new context that requires the same core idea rather than introducing an unrelated topic.

Quality notes

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